Brain Tumour Magnetic Resonance Image Classification with Perceptual-Hash Deduplication: A Comparative Study Using Advanced Deep Learning Models

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ABSTRACT

Magnetic resonance imaging (MRI) is the standard investigation for brain tumours, but published deep learning accuracies are difficult to compare because studies differ in data, splitting and evaluation, and near-duplicate images from shared public sources leak between training and test partitions. The contribution to artificial intelligence is a controlled evaluation protocol: label-aware perceptual-hash deduplication before splitting, a group-aware split into five cross-validation folds and a held-out test set, an equal number of optimiser steps for every model, temperature scaling, fold ensembles with bootstrap confidence intervals and Holm-corrected McNemar tests, and attention and background-only shortcut audits. The engineering application is the selection of a deployable tumour detector and subtype classifier. Nine ImageNet-pretrained convolutional, transformer and hybrid backbones were benchmarked on binary detection and three-way subtype classification. Deduplication reduced the binary detection pool from 3,253 to 1,185 images and the subtype classification pool from 5,828 to 4,730, and curation ran at about 85 images per second on four CPU cores. On 171 binary detection test images, CoAtNet-0, EfficientNetB0 and ViT-B/16 reached 99.42% accuracy (95% confidence interval 98.25–100%), and no pair of models differed significantly. The subtype classification grouping left only 45 test images, on which six models were error-free, so they cannot rank the models. Models retrained with the head masked out still exceeded the majority-class rate for 7 of 9 binary detection and all 9 subtype classification backbones, reaching 83.04% and 97.78% accuracy, so these public collections carry label information outside the brain. EfficientNetB0 matched the binary detection leaders with 4.01 million parameters.

1. Introduction

Brain tumours constitute the abnormal proliferation of cells either in the brain or the central nervous system, and they are classified by their cell of origin and molecular features (Louis et al., 2021). According to the Central Brain Tumor Registry of the United States, 453,623 primary brain and other central nervous system tumours were diagnosed in the United States between 2016 and 2020, an average of approximately 90,700 cases per year (Ostrom et al., 2023). Three major forms of primary brain tumours are studied in this paper. Glioma arises from glial cells and accounts for 26.3% of all primary brain and other central nervous system tumours in the same registry (Ostrom et al., 2023). Meningioma develops in the meningeal layers that cover the brain and is usually benign (Ostrom et al., 2023). Pituitary tumours grow in the pituitary gland (Bauer et al., 2013).

Magnetic resonance imaging (MRI) is the main non-invasive technique for brain tumour detection and characterisation (Villanueva-Meyer et al., 2017). Compared with computed tomography, MRI gives better soft tissue contrast without ionising radiation, and it acquires images with different types of contrast, called sequences, such as T1-weighted, contrast-enhanced T1-weighted, T2-weighted and fluid-attenuated inversion recovery (FLAIR), which together show tumour location, extent, margins and surrounding oedema (Villanueva-Meyer et al., 2017). An MRI study is a stack of two-dimensional images called slices. Reading these studies by hand is time-consuming and subject to variability between and within readers (Bauer et al., 2013), which motivates automated analysis.

Deep learning is a family of machine learning methods that learn layered representations of data directly from raw inputs (LeCun et al., 2015), and it has transformed medical image analysis over the last decade (Litjens et al., 2017). Its main tool for images is the convolutional neural network (CNN), which slides small learned filters over an image to detect local patterns and stacks such layers to build increasingly abstract features (LeCun et al., 1998). CNN architectures such as ResNet (He et al., 2016), VGG (Simonyan and Zisserman, 2015), DenseNet (Huang et al., 2017), Inception (Szegedy et al., 2016), EfficientNet (Tan and Le, 2019) and MobileNet (Howard et al., 2019) set the benchmarks of natural image classification. The transformer, introduced for natural language processing, instead relates every part of its input to every other part through a mechanism called self-attention (Vaswani et al., 2017). Vision

transformers such as ViT (Dosovitskiy et al., 2021) and the Swin Transformer (Liu et al., 2021), and hybrids such as CoAtNet (Dai et al., 2021), apply this mechanism to images, which may help to relate a tumour to the surrounding anatomy (Khan et al., 2022; Shamshad et al., 2023). All of these networks are usually pretrained on ImageNet, a collection of more than a million labelled natural photographs (Deng et al., 2009; Russakovsky et al., 2015), and then fine-tuned on the target task. The pretrained network without its final classification layer is called a backbone.

Although many studies report high accuracy for individual architectures, few compare several architectural families under the same experimental settings. Each study uses its own data, preprocessing, augmentation and evaluation, so it is impossible to tell whether an accuracy of 99% on a small, homogeneous collection is better than 97% on a larger, heterogeneous one. Comparisons are further distorted by data leakage, in which information from the test images reaches the model during training (Kapoor and Narayanan, 2023). Public brain tumour collections are frequently assembled from shared sources and carry no patient information, so duplicates must be removed before the data are split (Fateh et al., 2026). Otherwise the same image, or a near-identical copy, can appear in both the training and the test partition, a problem documented for combined public chest imaging collections (Roberts et al., 2021). A model that has seen an image during training is not tested on new data when it meets that image again, and its reported accuracy overstates its ability to generalise. Hashing can find near-identical copies, but it cannot find adjacent slices of the same patient, which look different but are not statistically independent. A further risk is shortcut learning, in which a model bases its prediction on a spurious cue, such as the image border or the scanner, rather than on the pathology (Geirhos et al., 2020; Zech et al., 2018; DeGrave et al., 2021).

Reliable automated detection could support faster preliminary assessment, more consistent interpretation and relief of radiologist workload (Beam and Kohane, 2018; De Fauw et al., 2018). This paper studies two tasks with the same pipeline. *Binary detection* decides whether an MRI slice contains a tumour, and *subtype classification* assigns a tumour-positive slice to glioma, meningioma or pituitary tumour. The two tasks are kept separate because subtype classification is only meaningful once a tumour-positive decision has been taken.

This paper makes four contributions. First, it trains and evaluates nine ImageNet-pretrained backbones from three architectural families with one code path, on identical splits and with an identical number of optimiser steps, so that the only difference between the rows of each results table is the architecture. Second, it controls leakage before any split is drawn, with label-aware perceptual-hash deduplication and a group-aware split whose leakage audit finds no test image near a development image, and it reports the measured processing time of this curation stage and how the deduplicated pool depends on the Hamming threshold. Third, it replaces single-run leaderboards with five-fold cross-validation, fold-ensemble test predictions, bootstrap confidence intervals, exact McNemar tests with Holm correction, calibration analysis and measured latency and throughput. Fourth, it audits every model for reliance on image regions outside the head, through attention maps and by retraining each model on background-only images.

Section 2 reviews existing studies, Section 3 describes the data, deduplication, splitting, models, training and evaluation, Section 4 presents the results, and Section 5 concludes. The supplementary material is a separate document whose figures and tables are numbered F1, F2, … and T1, T2, …. Section S1 of it holds the sample images and acquisition notes, and Section S2 the deduplication threshold sweep, the cross-label pairs, the group and leakage audit and the augmentation examples. Section S3 gives the complete training settings and evaluation procedures, and Section S4 the training curves and per-run results of every fold model. Section S5 holds the binary detection result table and the confusion matrices, receiver operating characteristic curves, McNemar tests and distance analysis of both tasks. Section S6 holds the reliability diagrams, the summary efficiency table and the accuracy–latency plots, and Section S7 the per-class attention table, the attention, occlusion and Grad-CAM maps of every model, and the masking and shortcut test results.

2. Literature Review

Early work on automated brain tumour recognition combined hand-engineered features with machine learning classifiers. Such work used texture descriptors such as the grey level co-occurrence matrix and local binary patterns, wavelet transforms, shape features, and classifiers such as the support vector machine (SVM), random forests and k-nearest neighbours (Bauer et al., 2013; Gordillo et al., 2013). Zacharaki et al. (2009) used texture and shape features and an SVM to separate metastases from gliomas with 85% accuracy and high-grade from low-grade tumours with 88%, and Cheng et al. (2015) reached 91.28% subtype classification accuracy on 3,064 MRI slices with intensity histogram, co-occurrence and bag-of-words features of an augmented, ring-partitioned tumour region and an SVM.

Hand-engineered features, however, must be redesigned for each problem and generalise poorly across datasets (Litjens et al., 2017).

The success of AlexNet in 2012 (Krizhevsky et al., 2017) brought deep learning to medical imaging (Litjens et al., 2017). He et al. (2016) introduced residual connections, which let very deep networks be trained by adding each block's input to its output. Fine-tuning a pretrained ResNet34, Talo et al. (2019) separated normal from abnormal brain MR images with 100% five-fold accuracy on 613 images, and Seetha and Raja (2018) reported 97.5% training accuracy for tumour versus non-tumour classification with an ImageNet-pretrained CNN of which only the last layer was trained. Simonyan and Zisserman (2015) showed that depth built from small 3×3 convolutional filters improves accuracy, and their VGG16 has about 138 million parameters. Rehman et al. (2020) reached 98.69% subtype classification accuracy on the 3,064 slices of Cheng et al. (2015) by fine-tuning it, and Khan et al. (2020) reached 97.8% on BraTS 2015 with VGG16 and VGG19 features and an extreme learning machine. Huang et al. (2017) connected each layer of a dense block to all preceding layers, and Noreen et al. (2020) reached 99.51% subtype classification accuracy on the same 3,064 slices by concatenating features from several dense blocks of DenseNet201. Szegedy et al. (2016) used parallel convolutions of different kernel sizes to learn multi-scale features. The same concatenation of Inception-v3 features reached 99.34% (Noreen et al., 2020), and features of GoogLeNet, the first Inception network, gave a mean accuracy of 98% under patient-level five-fold cross-validation (Deepak and Ameer, 2019). Ghassemi et al. (2020) instead pretrained a six-layer CNN as the discriminator of a generative adversarial network before training it on the 3,064 slices. Tan and Le (2019) scaled network depth, width and input resolution together from a baseline found by neural architecture search, obtaining better accuracy with far fewer parameters, and Howard et al. (2019) designed lightweight networks for mobile devices.

Dosovitskiy et al. (2021) adapted the transformer to images by treating an image as a sequence of 16×16 pixel patches. Their ViT matched CNNs on ImageNet when pretrained on very large datasets but lacks the built-in locality of convolution, which makes it data-hungry. Liu et al. (2021) restricted attention to shifted local windows in a hierarchy of scales, so that cost grows linearly with image size. Dai et al. (2021) stacked convolution stages followed by attention stages in the hybrid CoAtNet, combining the inductive bias of convolution with the global receptive field of attention. Shamshad et al. (2023) survey the adoption of transformers across medical imaging. For brain tumour MRI, Tummala et al. (2022) reported 98.7% test accuracy for subtype classification with an ensemble of four ViT variants on the 3,064-slice Cheng collection, and Asiri et al. (2024) reported 97% for a Swin Transformer on a 2,870-image collection with four classes, glioma, meningioma, pituitary tumour and no tumour.

The reliability of these benchmarks has itself come under scrutiny. Kapoor and Narayanan (2023) documented data leakage as a recurring cause of over-optimistic results in machine-learning-based science across 17 fields. Fateh et al. (2026), curating a widely used Kaggle aggregate of the Cheng, Bhuvaji and Br35H collections into the BRISC dataset, removed exact and near-duplicates before splitting to avoid leakage, and noted that the original release records no patient, sequence or slice information. In chest radiography, models have been shown to exploit hospital-specific markers and image borders instead of pathology (Zech et al., 2018; DeGrave et al., 2021). Table 1 summarises the brain tumour studies above that report a held-out or cross-validated accuracy for a named model.

Table 1
Accuracies reported by existing brain tumour MRI studies. Subtype classification is the three-way glioma, meningioma and pituitary task, and the 3,064 images are the collection of Cheng et al. (2015).

Study	Model	Dataset size	Task	Accuracy
Talo et al. (2019)	ResNet34	613	Normal vs abnormal	100.00%
Cheng et al. (2015)	SVM + features	3,064	Subtype classification	91.28%
Rehman et al. (2020)	VGG16	3,064	Subtype classification	98.69%
Noreen et al. (2020)	DenseNet201	3,064	Subtype classification	99.51%
Noreen et al. (2020)	Inception-v3	3,064	Subtype classification	99.34%
Deepak and Ameer (2019)	GoogLeNet	3,064	Subtype classification	98.00%
Tummala et al. (2022)	ViT ensemble	3,064	Subtype classification	98.70%
Asiri et al. (2024)	Swin Transformer	2,870	Four classes, incl. no tumour	97.00%

Several gaps remain. Most studies compare no more than two or three models, transformers are under-represented, and splits, preprocessing and evaluation differ between studies. Few studies remove duplicate and near-duplicate images

from combined collections, and source code and exact splits are rarely released. Results are commonly quoted from a single run without confidence intervals, significance tests, per-class sensitivity and specificity, calibration or measured inference cost, and whether a model relies on the anatomy or on the framing of the image is rarely tested. This paper addresses each of these gaps.

3. Methods and Materials

The workflow in Figure 1 is described stage by stage in Sections 3.1 to 3.5: the data, the preprocessing and deduplication, the group-aware split, the backbones, and training and evaluation. Both binary detection and subtype classification are run by the same notebook code, which differs only in the data sources and the number of output classes.

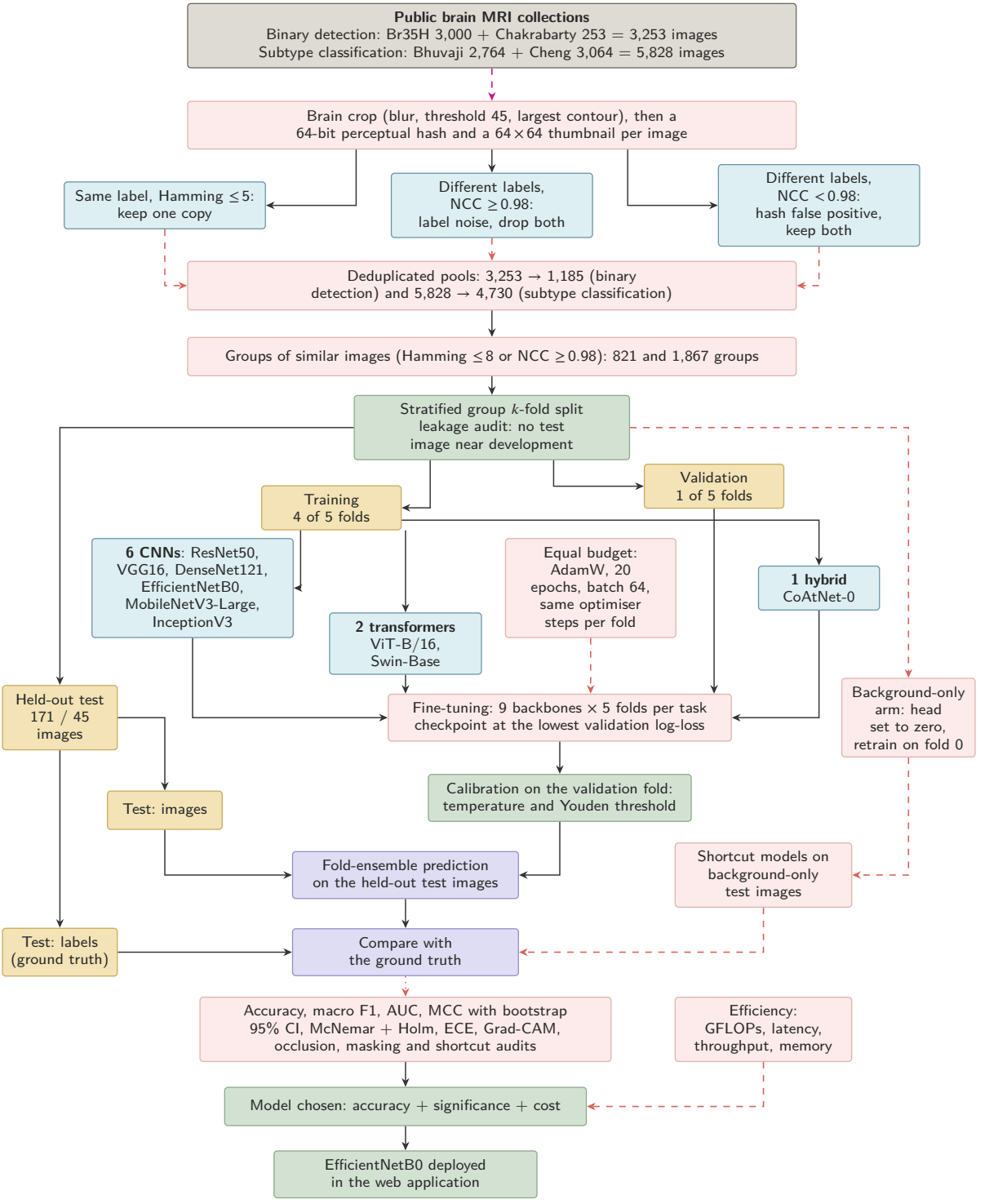


Figure 1: Overall methodology, run identically for binary detection and subtype classification. Deduplication and grouping precede every split, so no group of similar images crosses the held-out test set and the development folds. Each backbone is trained on every fold with the same number of optimiser steps, calibrated on its own validation fold and evaluated on the test set as a five-fold ensemble. Dashed arrows mark the curation outcomes, the background-only shortcut arm and the cost measurements that feed model selection.

3.1. Data Collection

Table 2 describes the four public collections used. For binary detection, *Brain MRI Images for Brain Tumor Detection* (Chakrabarty, 2019) and *Br35H: Brain Tumor Detection 2020* (Hamada, 2025) were combined into a pool of 3,253 images, 1,655 tumour-positive and 1,598 tumour-negative. The negative class consists of images that the dataset curators labelled as not containing a tumour. Neither source certifies that these images are radiologically normal, so the class is called tumour-negative rather than normal.

Table 2
Provenance of the source collections: images per class, native resolution as the number of distinct image sizes and the most common (modal) size with its share, the median fraction of the image kept by the brain crop, and the images removed as duplicates. Bhuvaji removals are recorded for both folders together.

Source	Images per class			Total	Sizes	Modal size (share)	Crop kept	Removed
Binary detection								
	Tumour	Tumour-neg.						
Br35H	1,500	1,500		3,000	291	225×225 (10.9%)	0.787	2,021
Chakrabarty	155	98		253	184	225×225 (7.1%)	0.783	47
Total	**1,655**	**1,598**		**3,253**				**2,068**
Subtype classification								
	Glioma	Meningioma	Pituitary					
Bhuvaji, training	826	822	827	2,475	137	512×512 (93.1%)	0.671	608
Bhuvaji, testing	100	115	74	289	184	512×512 (11.1%)	0.846	
Cheng	1,426	708	930	3,064	2	512×512 (99.5%)	0.635	486
Total	**2,352**	**1,645**	**1,831**	**5,828**				**1,094**

For subtype classification, the *Brain Tumor Classification (MRI)* collection (Bhuvaji et al., 2020), whose training and testing folders were merged and whose tumour-negative folder was excluded, was combined with the collection of Cheng (2017), 3,064 contrast-enhanced T1-weighted slices from 233 patients (Cheng et al., 2015) in the JPEG conversion distributed on Kaggle (MoundirB, 2025), into a pool of 5,828 images. No source records patient identifiers, and only the Cheng component records its MRI sequence (Cheng et al., 2015), so both pools are mixtures of unknown sequence composition. Sample images with their brain crops and head masks are shown in Supplementary F1.

3.2. Preprocessing and Deduplication

3.2.1. Brain cropping

Every image is converted to greyscale, smoothed with a 5×5 Gaussian filter to suppress noise, and thresholded at intensity 45 to separate the head from the dark background. The outer contours of the resulting binary mask are traced by border following (Suzuki and Abe, 1985), and the image is cropped to the bounding box of the largest one. This is the crop applied by the web application that deploys the models, so training, evaluation and deployment see the same framing. The crops are cached once at 320×320 pixels and resized to the input size of each backbone, and the median share of the image that the crop keeps is given for each source in Table 2.

3.2.2. Label-aware deduplication

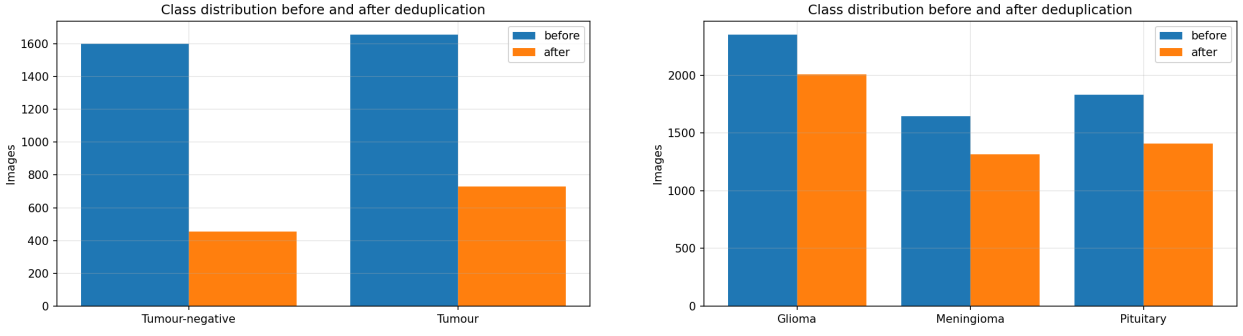
A perceptual hash summarises the visual content of an image in a short binary code, so that visually similar images receive similar codes even after resizing or recompression (Monga and Evans, 2006). The difference between two codes is measured by the Hamming distance, the number of bit positions in which they differ (Hamming, 1950). The notebooks use the discrete cosine transform (DCT) hash of the ImageHash library. Each original image, before cropping, is converted to greyscale and resized to 32×32 pixels, and the 8×8 lowest-frequency coefficients of its two-dimensional DCT are compared with their median, which gives a 64-bit code. A 64×64 greyscale thumbnail of every original image is stored as well.

Deduplication runs within each label, over the images in a fixed folder order. An image is removed when an image of the same label that has already been kept lies within Hamming distance 5 of it, and it is kept otherwise. Each kept image therefore stands for the near-copies removed against it. The removed images are not formed into explicit duplicate

sets, however, and two images removed against the same kept image may be more than 5 bits apart. Pairs within this distance that carry different labels were compared with the normalised cross-correlation (NCC) of their thumbnails, the correlation of their mean-subtracted pixel intensities, which lies between −1 and 1 and reaches 1 only for images that differ at most in brightness and contrast (Zitová and Flusser, 2003). At NCC $\geq$ 0.98 the pair is the same image filed under two labels, and both copies were excluded as label noise. Below that value the pair is a hash false positive, and both images were kept. As Table 3 and Figure 2 show, the pass removed 63.57% of the binary detection pool, which contained no cross-label pair, and 18.84% of the subtype classification pool. In the subtype classification pool, 8 of 31 cross-label pairs were label noise and flagged 10 images, 6 of which were already removed as duplicates. The cross-label pairs (Supplementary F2) and the per-source breakdown (Section S2) are in the supplementary material.

Table 3
Deduplication outcome per class at Hamming tolerance 5. Label noise counts only the label-noise images not already removed as duplicates: of the 10 subtype classification images in label-noise pairs (2 glioma, 8 meningioma), 6 meningioma images are among the duplicates.

Class	Before	Duplicates	Label noise	After	Removed (%)
Binary detection					
Tumour-negative	1,598	1,143	0	455	71.53
Tumour	1,655	925	0	730	55.89
Total	**3,253**	**2,068**	**0**	**1,185**	**63.57**
Subtype classification					
Glioma	2,352	342	2	2,008	14.63
Meningioma	1,645	329	2	1,314	20.12
Pituitary	1,831	423	0	1,408	23.10
Total	**5,828**	**1,094**	**4**	**4,730**	**18.84**



(a) Binary detection.

(b) Subtype classification.

Figure 2: Class distribution before and after deduplication at Hamming tolerance 5. Images kept at every other tolerance are listed in Supplementary T1.

3.2.3. Choice of the Hamming threshold

The threshold of 5 bits was carried over from an earlier version of this study. It was not derived from a published recommendation or from an experiment of this study, and it should not be read as an established value. Hamming (1950) defines the distance and Monga and Evans (2006) evaluate perceptual hashing in general, but neither gives a threshold for brain MRI slices. Two properties of the hashes recorded by the notebooks show what the choice does and does not decide (Supplementary T1).

- **Thresholds 4 and 5 are identical.** Because the 64 coefficients are split at their own median, every hash in both pools has exactly 32 bits set. Two such codes differ in 64 minus twice the number of set bits they share, so every pairwise distance is even, and none of the 5.3 million binary detection and 17.0 million subtype classification

pairs has an odd distance. A threshold of 5 therefore keeps exactly the same pool as a threshold of 4, and the real alternatives are 2 and 6.

- **Cross-label matches grow quickly above 4.** Images with different labels cannot be copies of one image unless a label is wrong, so the number of cross-label pairs within the threshold indicates how often distinct images fall inside it. Up to distance 4 there are no such pairs in binary detection and 31 in subtype classification, 23 of which the thumbnails show to be different images. At distance 6 there would be 1 and 253. Raising the threshold from 4 to 6 would also remove a further 183 binary detection and 806 subtype classification images. Lowering it to 2 would keep 310 and 603 more.

A threshold of 4 or 5 is therefore a conservative point at which cross-label matches are still rare. Nothing in the data establishes that every same-label pair within it is a copy of one image, or that no copy lies beyond it, because no set of duplicate pairs confirmed by inspection exists for these collections. The sweep measures only how many images are kept, not how the threshold changes the split or the test results. Before the threshold can be treated as validated, a sensitivity analysis should repeat the deduplication, grouping, training and evaluation at thresholds of 2, 4, 6 and 8, and compare the resulting pools against a sample of pairs labelled as duplicate or distinct by visual inspection.

3.3. Group-Aware Splitting

Cross-validation estimates how well a model generalises by training it repeatedly on part of the data and evaluating it on the rest (Stone, 1974). Here the development images are divided into five folds, and each fold model is trained on four folds and validated on the fifth, while a separate held-out test set is never used for training or model selection. Because deduplication does not remove correlated but distinct images such as adjacent slices, kept images within Hamming distance 8 of each other or with thumbnail NCC ≥ 0.98 were first joined into groups (connected components of the similarity graph), a proxy for patient-level grouping. Stratified group k-fold sampling (Pedregosa et al., 2011), which keeps every group in one partition while preserving class proportions as far as possible, then held out one of seven group folds, about 14% of the images, as the test set and divided the rest into the five development folds. A leakage audit then found no test image within Hamming distance 5 or 8 or at NCC ≥ 0.98 of any development image in either task (Supplementary F3). The binary detection split is balanced (Table 4), but one subtype classification group held 1,431 images, and because a group cannot be divided, the subtype classification development folds range from 154 to 2,216 images and the held-out test set holds only 45 images, a limitation discussed in Section 4.3.

Table 4
Images per partition after deduplication and grouping. Development images are divided into five group-aware folds, whose sizes are given in the last column.

Class	Development	Test	Fold sizes (0–4)
Binary detection (821 groups, the largest of 7 images)			
Tumour-negative	391	64	
Tumour	623	107	
Total	**1,014**	**171**	184, 211, 207, 210, 202
Subtype classification (1,867 groups, the largest of 1,431 images)			
Glioma	2,007	1	
Meningioma	1,301	13	
Pituitary	1,377	31	
Total	**4,685**	**45**	154, 742, 697, 876, 2,216

3.4. Model Development

Nine architectures from three families were built from one model registry, so that a model name means the same implementation and pretrained checkpoint in both tasks. Eight are `torchvision` models of PyTorch (Paszke et al., 2019) loaded with their ImageNet weights (Deng et al., 2009; Russakovsky et al., 2015), and CoAtNet-0 is the `timm` model `coatnet_0_rw_224`, since `torchvision` has no CoAtNet. In every model the final linear layer is replaced by one with one output for binary detection or three for subtype classification, and the auxiliary classifier of InceptionV3 is removed so that no model receives an extra loss term. Six of the models are CNNs. ResNet50 is a 50-layer residual network

whose bottleneck blocks add their input to their output through skip connections (He et al., 2016). VGG16 has 13 convolutional layers of 3×3 filters followed by three fully connected layers (Simonyan and Zisserman, 2015). In DenseNet121, each layer of a dense block receives the concatenated feature maps of all preceding layers (Huang et al., 2017). EfficientNetB0 is the baseline network found by neural architecture search from which the EfficientNet family is obtained by compound scaling (Tan and Le, 2019). MobileNetV3-Large is built from inverted residual blocks with depth-wise separable convolutions (Sandler et al., 2018) and tuned for mobile devices by architecture search (Howard et al., 2019). InceptionV3 runs factorised convolutions of different sizes in parallel branches and takes 299×299 inputs (Szegedy et al., 2016). The two transformers are ViT-B/16, which splits the image into 16×16 patches and processes them as a token sequence with a 12-layer transformer encoder (Dosovitskiy et al., 2021), and Swin-Base, which computes self-attention inside shifted local windows of a four-stage hierarchy (Liu et al., 2021). The hybrid CoAtNet-0 uses depth-wise convolution in its early stages and relative self-attention in its later stages (Dai et al., 2021). The parameter count of every model, measured on the instantiated network, is reported with its measured cost in Section 4.5.

3.5. Training Configuration and Evaluation Protocol

All runs used two NVIDIA Tesla T4 graphics processing units on Kaggle with PyTorch 2.11.0 (Paszke et al., 2019) and identical settings for every model and both tasks (Supplementary T2): crops normalised with the ImageNet channel means and standard deviations, training augmentation by horizontal flipping, rotation of up to $\pm 10°$ and brightness and contrast jitter of up to 20% (Supplementary F4), 20 epochs with an effective batch of 64 images made up by gradient accumulation where a model did not fit on one GPU, so that every model took the same number of optimiser steps on a given fold, the AdamW optimiser (Loshchilov and Hutter, 2019) with learning rate 3×10^{-4} and weight decay 0.01, a one-epoch linear warm-up followed by cosine decay (Loshchilov and Hutter, 2017), half-precision training (Micikevicius et al., 2018), gradient clipping at norm 1.0 (Pascanu et al., 2013), binary cross-entropy for binary detection and cross-entropy for subtype classification, and, after each epoch, retention of the checkpoint with the lowest validation log-loss, a strictly proper scoring rule (Gneiting and Raftery, 2007). Each backbone was also trained once on fold 0 with the head set to zero for the shortcut test, giving 54 runs per task, all of which completed (details in Section S3).

Calibration and thresholds. For every run a temperature that rescales the model's outputs was fitted on its own validation fold by minimising the negative log-likelihood (Guo et al., 2017), and for binary detection a decision threshold was chosen there by Youden's J statistic, the threshold that maximises sensitivity plus specificity minus one (Youden, 1950).

Cross-validation and fold ensembles. The five fold models of each backbone were scored individually on the held-out test set and also averaged into one fold-ensemble prediction per test image, whose accuracy, macro F1 and area under the receiver operating characteristic curve (AUC) received percentile bootstrap 95% confidence intervals from 2,000 resamples of the test images (Efron, 1979).

Statistical comparison. Every pair of fold ensembles was compared with an exact McNemar test on the images that exactly one of the two misclassifies (McNemar, 1947), with p-values adjusted for the 36 comparisons by the Holm procedure (Holm, 1979).

Attention and shortcut analysis. For every backbone, gradient-weighted class activation maps (Grad-CAM) (Selvaraju et al., 2017), attention rollout for ViT-B/16 (Abnar and Zuidema, 2020), occlusion sensitivity (Zeiler and Fergus, 2014), inference with the head or the background masked out by an Otsu head mask (Otsu, 1979), and retraining on background-only images (Geirhos et al., 2020) measured how much each model relies on the head, as defined in Section 4.7.

Efficiency. Parameters, giga floating-point operations (GFLOPs) per image, single-image latency, batched throughput and peak training memory were measured for every model on one T4 under half-precision autocasting. The wall-clock time of the curation stage, which decodes, hashes and crops every image in parallel worker processes, one per CPU core, was logged by the notebooks.

4. Results and Discussion

Section 4.1 reports five-fold cross-validation, Sections 4.2 and 4.3 the fold-ensemble test results of binary detection and subtype classification, Section 4.4 calibration, Section 4.5 efficiency, Section 4.6 the processing time and scalability

of the curation stage, Section 4.7 attention and shortcut analyses, and Section 4.8 the comparison with published work. The per-run tables and per-model figures behind each section are in Sections S4 to S7 of the supplementary material.

4.1. Five-Fold Cross-Validation Spread of the Fold Models

A fold model is one of the five models trained per backbone, each on four of the five development folds, and Table 5 reports the mean and standard deviation of their test results, where accuracy is the share of test images classified correctly, macro F1 is the per-class harmonic mean of precision and sensitivity averaged over classes, and AUC is the probability that a randomly chosen positive image is scored above a randomly chosen negative one (Hanley and McNeil, 1982; Fawcett, 2006). In binary detection, mean accuracy ranges from 95.91% for VGG16 to 97.89% for InceptionV3, and the standard deviations across folds, from 0.32 to 2.20 percentage points, are of the same size as most gaps between neighbouring models. CoAtNet-0, for example, scores between 94.74% and 98.83% on its five folds, so the choice of training folds matters as much as the choice of backbone. The binary detection training curves and the temperature, threshold and test result of every binary detection run are given in Supplementary F5 and Supplementary T3.

In subtype classification, every model averages between 97.33% and 99.56% accuracy, but each fold model is scored on the same 45 test images, so a single image changes accuracy by 2.22 percentage points, and the uneven folds give each fold model between 780 and 1,420 optimiser steps. The subtype classification training curves and per-run results are given in Supplementary F6 and Supplementary T4.

Table 5
Test results of the individual fold models: mean ± standard deviation over the five folds of each backbone, after temperature scaling. Binary detection is evaluated on 171 test images and subtype classification on 45. Rows are ordered by binary detection accuracy.

	Binary detection			Subtype classification		
Model	Accuracy (%)	Macro F1 (%)	AUC	Accuracy (%)	Macro F1 (%)	AUC
InceptionV3	97.89 ± 0.32	97.75 ± 0.34	0.9966 ± 0.0042	99.56 ± 0.99	97.51 ± 5.57	1.0000 ± 0.0000
Swin-Base	97.78 ± 1.62	97.64 ± 1.72	0.9974 ± 0.0026	99.56 ± 0.99	97.51 ± 5.57	1.0000 ± 0.0000
EfficientNetB0	97.78 ± 0.49	97.62 ± 0.52	0.9968 ± 0.0034	97.33 ± 3.65	90.61 ± 9.43	0.9989 ± 0.0017
CoAtNet-0	97.54 ± 1.72	97.39 ± 1.83	0.9966 ± 0.0038	98.67 ± 1.22	94.67 ± 6.53	1.0000 ± 0.0000
ViT-B/16	97.54 ± 0.96	97.37 ± 1.04	0.9970 ± 0.0013	99.11 ± 1.22	95.02 ± 6.82	0.9995 ± 0.0011
DenseNet121	97.19 ± 0.76	96.98 ± 0.82	0.9927 ± 0.0048	99.11 ± 1.22	97.15 ± 5.42	0.9998 ± 0.0004
ResNet50	96.84 ± 1.35	96.60 ± 1.46	0.9876 ± 0.0105	98.67 ± 1.22	92.69 ± 6.68	0.9998 ± 0.0004
MobileNetV3-Large	96.37 ± 2.20	96.16 ± 2.30	0.9946 ± 0.0033	97.78 ± 0.00	87.56 ± 0.00	0.9998 ± 0.0004
VGG16	95.91 ± 0.83	95.65 ± 0.88	0.9943 ± 0.0023	97.33 ± 2.43	88.46 ± 7.30	0.9989 ± 0.0021

4.2. Binary Detection Results

A fold ensemble averages the calibrated probabilities of a backbone's five fold models into one prediction per test image, and it is evaluated with sensitivity (the share of tumour images detected), specificity (the share of tumour-negative images correctly cleared), their mean, called balanced accuracy, and the Matthews correlation coefficient (MCC), a correlation between predicted and true labels that remains informative when classes are imbalanced (Matthews, 1975; Chicco and Jurman, 2020). On the 171 binary detection test images, CoAtNet-0, EfficientNetB0 and ViT-B/16 each misclassified one image, an accuracy of 99.42% with a bootstrap 95% confidence interval of 98.25% to 100% (Supplementary T5). Their single errors differ in kind: CoAtNet-0 missed one tumour, whereas EfficientNetB0 and ViT-B/16 detected every tumour and raised one false alarm (Table 6). Swin-Base followed at 98.83% with two missed tumours, then InceptionV3 and ResNet50 at 98.25%, and DenseNet121, MobileNetV3-Large and VGG16 at 97.08%, and every AUC lies between 0.9974 and 0.9997.

Table 6
Per-class test metrics of the fold ensembles. TP, FP and FN are true positives, false positives and false negatives of each class. Binary detection support: 64 tumour-negative, 107 tumour. Subtype classification support: 1 glioma, 13 meningioma, 31 pituitary.

Model	Class	TP	FP	FN	Precision	Sensitivity	Specificity
Binary detection							
ResNet50	Tumour-negative	63	2	1	0.9692	0.9844	0.9813
	Tumour	105	1	2	0.9906	0.9813	0.9844
VGG16	Tumour-negative	62	3	2	0.9538	0.9688	0.9720
	Tumour	104	2	3	0.9811	0.9720	0.9688
DenseNet121	Tumour-negative	60	1	4	0.9836	0.9375	0.9907
	Tumour	106	4	1	0.9636	0.9907	0.9375
EfficientNetB0	Tumour-negative	63	0	1	1.0000	0.9844	1.0000
	Tumour	107	1	0	0.9907	1.0000	0.9844
MobileNetV3-Large	Tumour-negative	62	3	2	0.9538	0.9688	0.9720
	Tumour	104	2	3	0.9811	0.9720	0.9688
InceptionV3	Tumour-negative	62	1	2	0.9841	0.9688	0.9907
	Tumour	106	2	1	0.9815	0.9907	0.9688
ViT-B/16	Tumour-negative	63	0	1	1.0000	0.9844	1.0000
	Tumour	107	1	0	0.9907	1.0000	0.9844
Swin-Base	Tumour-negative	64	2	0	0.9697	1.0000	0.9813
	Tumour	105	0	2	1.0000	0.9813	1.0000
CoAtNet-0	Tumour-negative	64	1	0	0.9846	1.0000	0.9907
	Tumour	106	0	1	1.0000	0.9907	1.0000
Subtype classification							
ResNet50, DenseNet121, InceptionV3, ViT-B/16, Swin-Base, CoAtNet-0	Glioma	1	0	0	1.0000	1.0000	1.0000
	Meningioma	13	0	0	1.0000	1.0000	1.0000
	Pituitary	31	0	0	1.0000	1.0000	1.0000
VGG16, EfficientNetB0, MobileNetV3-Large	Glioma	1	1	0	0.5000	1.0000	0.9773
	Meningioma	12	0	1	1.0000	0.9231	1.0000
	Pituitary	31	0	0	1.0000	1.0000	1.0000

These differences are small relative to the test set. No pair of models differs significantly by the exact McNemar test after Holm correction, since every adjusted p-value is 1.0, and the bootstrap interval of the accuracy gap to CoAtNet-0 excludes zero only for VGG16 and MobileNetV3-Large, both 2.34 percentage points behind with an interval of 0.58 to 4.68 points. Accuracy also does not fall for test images that are furthest from the development pool: with nearest development images at Hamming distance 17 to 20, every model is correct on every image. The binary detection confusion matrices, receiver operating characteristic and precision–recall curves, McNemar p-values, accuracy by distance and results at the validation-tuned thresholds are given in Supplementary F7, F9, F10 and F11, Supplementary T6 and Section S5.

4.3. Subtype Classification Results

Subtype classification is scored with the same metrics, with precision, F1 and AUC macro-averaged, that is, computed for each subtype against the other two and averaged so that every subtype counts equally. On the 45 subtype classification test images, ResNet50, DenseNet121, InceptionV3, ViT-B/16, Swin-Base and CoAtNet-0 classified every image correctly (Table 7). VGG16, EfficientNetB0 and MobileNetV3-Large made the same single error, a meningioma image predicted as glioma, for an accuracy of 97.78%. Because the test set contains only one glioma image, that error halves glioma precision (Table 6) and pulls macro F1 down to 87.56%. Every model reaches an AUC of 1.0, no McNemar comparison is significant, and every bootstrap interval of the gap to the leader includes zero. The subtype classification confusion matrices, curves, McNemar p-values and accuracy by distance are given in Supplementary F8, F9, F10 and F11.

Table 7
Subtype classification: fold ensembles on the 45 held-out test images (1 glioma, 13 meningioma, 31 pituitary). Precision and F1 are macro-averaged, and AUC is the macro one-versus-rest value. Accuracy has a bootstrap 95% confidence interval.

Model	Accuracy (%) [95% CI]	Bal. acc. (%)	Prec. (%)	F1 (%)	MCC	AUC
ResNet50	100.00 [100, 100]	100.00	100.00	100.00	1.0000	1.0000
DenseNet121	100.00 [100, 100]	100.00	100.00	100.00	1.0000	1.0000
InceptionV3	100.00 [100, 100]	100.00	100.00	100.00	1.0000	1.0000
ViT-B/16	100.00 [100, 100]	100.00	100.00	100.00	1.0000	1.0000
Swin-Base	100.00 [100, 100]	100.00	100.00	100.00	1.0000	1.0000
CoAtNet-0	100.00 [100, 100]	100.00	100.00	100.00	1.0000	1.0000
VGG16	97.78 [93.33, 100]	97.44	83.33	87.56	0.9515	1.0000
EfficientNetB0	97.78 [93.33, 100]	97.44	83.33	87.56	0.9515	1.0000
MobileNetV3-Large	97.78 [93.33, 100]	97.44	83.33	87.56	0.9515	1.0000

This test set cannot rank the models. It contains 45 images, one of them glioma, because a single group of 1,431 images could not be divided between partitions (Section 3.3). Its class balance (2.2% glioma, 28.9% meningioma, 68.9% pituitary) is far from that of the pool, and the cross-validation results in Table 5 are scored on the same 45 images. The subtype classification results therefore show that every backbone separates these subtypes on images that are not near-copies of its training images, but they do not establish an ordering of the backbones, and the 100% accuracies should not be quoted as an estimate of subtype classification accuracy in general.

4.4. Calibration

A model is well calibrated when its predicted probabilities match observed frequencies, so that, for example, 90% of the images it scores at 0.9 belong to the predicted class (Guo et al., 2017). The expected calibration error (ECE) measures the gap by sorting predictions into 15 equal-width confidence bins and averaging the difference between confidence and accuracy over the bins, weighted by their size (Pakdaman Naeini et al., 2015), and temperature scaling divides the model's outputs by a single fitted constant T to correct over- or under-confidence without changing any prediction (Guo et al., 2017). Table 8 gives the ECE of the fold ensembles before and after temperature scaling. Every model is reasonably calibrated, with ECE between 0.0146 and 0.0326 in binary detection and between 0.0044 and 0.0371 in subtype classification, but temperature scaling lowered the ECE only for ResNet50 and ViT-B/16 in binary detection and for CoAtNet-0 in subtype classification, and raised it for every other model. When a validation fold is classified almost perfectly, minimising its negative log-likelihood drives the temperature towards the lower bound of the search, $e^{-3} \approx 0.0498$, which sharpens the test probabilities: in binary detection fold 4 the fitted temperature of six models lies between 0.0498 and 0.0682, and for five of them that fold has the highest test log-loss of the model's five folds (Supplementary T3). Validation folds of a few hundred near-separable images are therefore too small to fit a temperature reliably. Reliability diagrams of every model are given in Supplementary F12.

Table 8
Expected calibration error of the fold ensembles on the held-out test set, after and before temperature scaling.

	Binary detection		Subtype classification	
Model	Scaled	Uncalibrated	Scaled	Uncalibrated
CoAtNet-0	0.0275	0.0207	0.0253	0.0259
DenseNet121	0.0252	0.0233	0.0215	0.0165
EfficientNetB0	0.0243	0.0231	0.0277	0.0263
InceptionV3	0.0228	0.0151	0.0088	0.0045
MobileNetV3-Large	0.0326	0.0146	0.0285	0.0204
ResNet50	0.0157	0.0173	0.0222	0.0125
Swin-Base	0.0152	0.0148	0.0109	0.0044
VGG16	0.0296	0.0179	0.0371	0.0355
ViT-B/16	0.0207	0.0222	0.0182	0.0114

4.5. Model Efficiency Comparison

Efficiency is measured by the number of learned parameters, the GFLOPs needed to classify one image, the single-image latency (time from input to prediction), the throughput (images classified per second when 32 are processed together), the training throughput and the peak GPU memory used in training. Table 9 gives the complete measurements. Cost varies by a factor of about 72, from 0.43 GFLOPs for MobileNetV3-Large to 30.9 GFLOPs for Swin-Base and VGG16 (Supplementary T7), while binary detection test accuracy varies by 2.34 percentage points and the differences are not significant. MobileNetV3-Large has the highest batched throughput, 2,301 images per second in binary detection, and EfficientNetB0 follows at 1,393 images per second with an accuracy equal to the leaders. Swin-Base has the lowest throughput at 198 images per second and the highest single-image latency at 36.3 ms. Single-image latency does not follow model size: VGG16, the largest model, has the lowest latency at 7.7 ms in binary detection, whereas DenseNet121 needs 23.6 ms despite its 6.95 million parameters, so neither parameter count nor GFLOPs predicts single-image latency on this hardware. Accuracy is plotted against latency in Supplementary F13.

Table 9
Efficiency measurements. The parameter count of each model, in millions, is given in brackets, and the three-output subtype classification heads add a few thousand parameters. Latency (ms) and throughput at batch size 32 (images/s) are mean ± pooled standard deviation over three repeats on one Tesla T4 under half-precision autocasting. Training throughput (images/s) and peak training memory (GB) are averaged over the folds. Training throughput is a lower bound because the two workers shared four CPU cores.

Model (parameters, M)	Latency (ms)	Throughput (img/s)	Training (img/s)	Peak memory (GB)
Binary detection				
ResNet50 (23.51)	9.57 ± 0.38	777.2 ± 6.1	235.2	5.62
VGG16 (134.26)	7.72 ± 0.42	386.5 ± 4.8	124.4	9.88
DenseNet121 (6.95)	23.58 ± 1.53	656.6 ± 22.9	185.4	4.48
EfficientNetB0 (4.01)	13.13 ± 0.55	1393.3 ± 115.8	136.4	3.09
MobileNetV3-Large (4.20)	10.55 ± 1.09	2301.1 ± 292.9	182.3	2.09
InceptionV3 (21.79)	17.63 ± 0.79	511.6 ± 5.8	155.8	5.85
ViT-B/16 (85.80)	10.95 ± 1.04	272.2 ± 2.8	93.0	5.96
Swin-Base (86.74)	36.29 ± 1.23	197.6 ± 2.9	62.9	6.45
CoAtNet-0 (26.67)	16.53 ± 1.07	513.1 ± 6.9	118.7	11.39
Subtype classification				
ResNet50 (23.51)	9.07 ± 0.36	716.8 ± 6.9	232.9	4.08
VGG16 (134.27)	8.02 ± 0.40	332.1 ± 6.2	130.5	9.87
DenseNet121 (6.96)	22.81 ± 1.22	607.7 ± 17.7	180.4	4.48
EfficientNetB0 (4.01)	12.81 ± 0.80	1265.7 ± 50.3	243.9	3.04
MobileNetV3-Large (4.21)	9.99 ± 0.92	2183.1 ± 234.4	355.2	2.08
InceptionV3 (21.79)	16.69 ± 1.25	438.2 ± 6.6	155.4	5.26
ViT-B/16 (85.80)	10.26 ± 1.10	241.0 ± 3.0	82.6	5.97
Swin-Base (86.75)	34.65 ± 1.51	173.1 ± 2.8	57.8	6.45
CoAtNet-0 (26.67)	15.88 ± 0.73	448.4 ± 6.3	131.1	10.42

4.6. Processing Time and Scalability of the Curation Stage

The curation stage was timed as a whole by the notebooks, on the four CPU cores of the Kaggle machine with one worker process per core, and Table 10 gives the measurements. Each worker decodes an image, computes its 64-bit hash and 64×64 thumbnail, crops it to the brain, resizes the crop to 320×320 pixels and computes its head mask. The 3,253 binary detection images took 39 s and the 5,828 subtype classification images 68 s, about 83 and 86 images per second, or about 12 ms of wall-clock time per image. The hash was not timed apart from the other steps, so these rates are lower bounds for hashing alone, and the time the hash itself takes is not known. The deduplication step that follows was not timed separately either. The notebook log, whose timestamps have a resolution of one second, places it within about 2 s for binary detection and 3 s for subtype classification, including the full matrix of pairwise Hamming distances, the label-aware pass, the cross-label check and the writing of its tables. Curation was therefore a small part of the 72-minute binary detection and 274-minute subtype classification notebook runs, which were dominated by training.

Table 10
Measured processing time of the curation stage on four CPU cores (Kaggle). Decode, hash and crop: time logged by the notebook for decoding, hashing, thumbnailing, brain cropping, resizing and head masking of every image, rounded to whole seconds by the log. Deduplication: upper bound from the one-second log timestamps around that step, not a dedicated timing. Distance matrix: number of one-byte entries of the pairwise Hamming distance matrix.

Task	Images	Decode, hash and crop (s)	Images/s	ms/image	Deduplication (s)	Distance matrix (million entries)
Binary detection	3,253	39	83	12.0	≤ 2	10.6
Subtype classification	5,828	68	86	11.7	≤ 3	34.0

The two parts of the stage scale differently with the number of images. Decoding, hashing and cropping treat every image independently, so their cost grows linearly with the number of images and divides among worker processes. The two runs agree with this, with similar rates for pools of 3,253 and 5,828 images, but two pool sizes on one machine are too few to measure the scaling itself. The comparison of hashes, in contrast, builds the full matrix of pairwise distances, whose size grows with the square of the number of images. It holds 10.6 million one-byte entries for binary detection and 34.0 million for subtype classification, but it would hold 10^{10} entries, 10 GB, for 100,000 images. Exhaustive comparison is thus adequate for pools of the size studied here, but not for a multi-institutional collection of hundreds of thousands of slices. Such a collection would need an index that returns all codes within the threshold without comparing every pair. Multi-index hashing performs this search exactly and ran in sub-linear time on uniformly distributed codes in the evaluation of Norouzi et al. (2014), which included up to one billion 64-bit codes. The hashes of MRI slices are not uniformly distributed, however, and such an index was not tested here. Each hash takes 8 bytes, so the codes of one million images occupy 8 MB and can be compared in one place.

On the measured results, the pipeline suits the curation of public collections of a few thousand images on an ordinary CPU, as a step taken once before any split. Its use for large, multi-institutional curation is plausible for the hashing, which is linear and parallel, but it is not demonstrated by these experiments. Such use would need an indexed comparison, a validated threshold (Section 3.2.3) and a test of whether one threshold transfers across scanners and institutions. The hash also finds only near-copies of an image, so repeated scans of one patient on different scanners would still need patient identifiers to be grouped.

4.7. Interpretability and Shortcut Analysis

Interpretability analysis asks which parts of an image a model uses, and shortcut analysis asks whether it can succeed with cues unrelated to the pathology (Geirhos et al., 2020). Every measurement below uses a head mask computed for each test image by Otsu thresholding (Otsu, 1979), morphological closing, selection of the largest connected component and hole filling, and Table 11 summarises them.

- **Grad-CAM energy outside the head.** Grad-CAM weights the feature maps of a network's last spatial stage by the gradient of the predicted class score and sums them into a heat map of the regions that raised that score (Selvaraju et al., 2017). The metric is the share of this map's total intensity (energy) that falls outside the head mask, compared with the share a uniform map would give, which equals the share of the image outside the head. The share is below the uniform value for most models and classes. ViT-B/16 is the exception in binary detection, with 42.6% of its Grad-CAM energy outside the head on tumour-negative images against 35.6% for a uniform map, and in subtype classification ViT-B/16 and CoAtNet-0 place 32.6% and 41.7% outside the head on the single glioma test image against 21.3% for a uniform map (Supplementary T8, Supplementary F14).

- **Attention rollout.** Attention rollout multiplies the attention matrices of all transformer layers to trace how much each image patch contributes to the final prediction (Abnar and Zuidema, 2020). For ViT-B/16 it places 42.1% of its weight outside the head on binary detection tumour-negative images, agreeing with Grad-CAM, and in subtype classification it places 35.0% and 35.8% outside the head on meningioma and pituitary images, above the uniform 26.8% and 26.7%, where Grad-CAM stays below them.

- **Occlusion sensitivity.** Occlusion sensitivity replaces a 32×32 patch, moved with stride 16, by the image mean and records the drop in the predicted-class probability (Zeiler and Fergus, 2014). Because it needs no access to the network's internals, it is computed the same way for every architecture on a common stratified sample of test

images, and its share outside the head ranges from 0.7% (VGG16) to 19.4% (EfficientNetB0) in binary detection and from 6.6% to 12.4% in subtype classification (Supplementary F15 and F16).

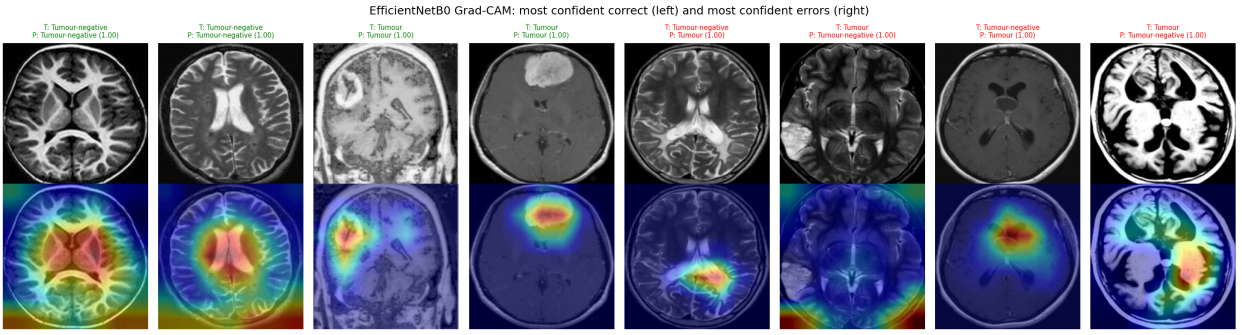
- **Head-only and background-only accuracy.** The masking test applies each fold model, unchanged, to test images whose background is set to zero (head-only) or whose head is set to zero (background-only), and reports the mean accuracy over folds. A model that depends on the head should keep its accuracy on head-only images and fall to chance on background-only images.

- **Shortcut accuracy.** The shortcut test retrains each backbone on background-only images and evaluates it on background-only test images. Its accuracy is compared with the majority-class rate, the accuracy of always predicting the most frequent class, by a one-sided exact binomial test with Holm correction (Holm, 1979), and accuracy significantly above that rate shows that the background carries label information a model can learn (Geirhos et al., 2020).

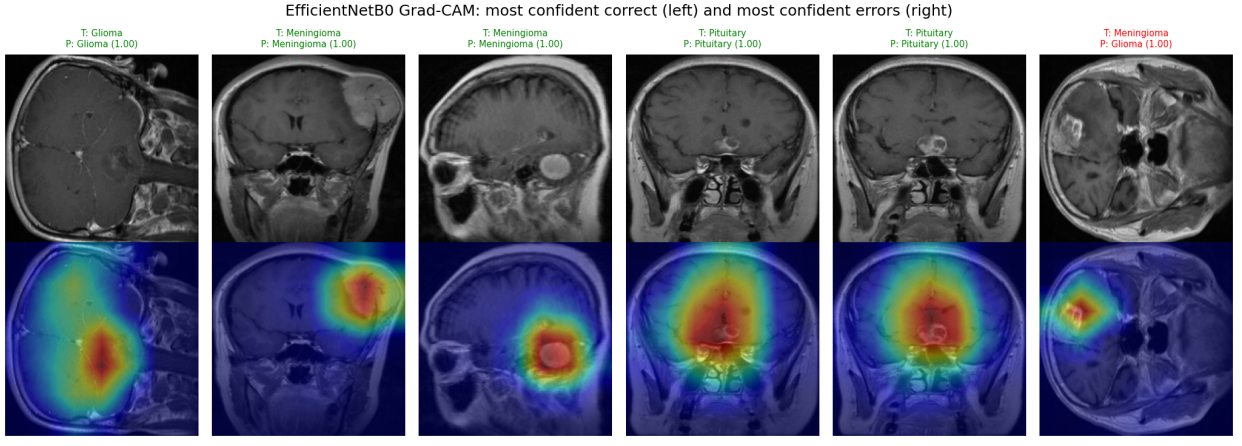
Table 11
Reliance on the head and on the background. Grad-CAM: share of map energy outside the head mask, the average of the per-class means over test images (uniform map: 0.34 binary detection, 0.25 subtype classification, averaged the same way). Occl.: share of occlusion sensitivity outside the head on 16 (binary detection) and 17 (subtype classification) test images. Head and Backgr.: mean fold accuracy on head-only and background-only test images. Shortcut: accuracy on background-only test images of a model trained on background-only images, with majority-class rates of 0.626 (binary detection) and 0.689 (subtype classification). An asterisk (*) marks accuracy above that rate after Holm correction.

	Binary detection					Subtype classification				
Model	Grad-CAM	Occl.	Head	Backgr.	Shortcut	Grad-CAM	Occl.	Head	Backgr.	Shortcut
ResNet50	0.100	0.116	0.960	0.492	0.813*	0.087	0.079	0.947	0.284	0.933*
VGG16	0.113	0.007	0.947	0.499	0.749*	0.087	0.079	0.942	0.240	0.978*
DenseNet121	0.183	0.135	0.944	0.518	0.795*	0.135	0.110	0.960	0.276	0.867*
EfficientNetB0	0.179	0.194	0.956	0.498	0.778*	0.064	0.066	0.956	0.133	0.889*
MobileNetV3-Large	0.112	0.111	0.937	0.533	0.678	0.077	0.071	0.951	0.338	0.933*
InceptionV3	0.147	0.167	0.945	0.657	0.795*	0.122	0.099	0.947	0.218	0.978*
ViT-B/16	0.335	0.155	0.957	0.450	0.655	0.207	0.107	0.987	0.116	0.911*
Swin-Base	0.172	0.182	0.968	0.517	0.830*	0.102	0.122	0.996	0.169	0.911*
CoAtNet-0	0.164	0.089	0.964	0.444	0.795*	0.231	0.124	0.982	0.156	0.822*

- **Grad-CAM maps.** Figure 3 shows the Grad-CAM maps of EfficientNetB0, the architecture deployed in the accompanying application, on the test images it predicted most confidently, correct predictions on the left and errors on the right, from the fold model with the lowest validation loss. On correctly detected tumours and on all five correctly classified subtype images, the warmest region lies on the visible lesion: the frontal mass and the coronal lesion in binary detection, and the meningioma, the glioma and the sellar pituitary tumours in subtype classification. On correctly cleared tumour-negative images the map covers the central brain and the ventricles, but it also lights up the lower image border, outside the head. The binary detection errors are instructive: on a tumour image with a visible right temporal lesion that the model called tumour-negative, the map ignores the lesion and peaks at the lower border, the kind of cue the shortcut test below shows these collections contain. These maps come from a single fold model, which makes more errors than the five-fold ensemble that misclassified only one binary detection image. The maps of the other eight backbones are given in Supplementary F17–F24.



(a) Binary detection.



(b) Subtype classification.

Figure 3: Grad-CAM maps of EfficientNetB0, the deployed architecture, on its most confident correct (green titles) and most confident wrong (red titles) test predictions, from the fold with the lowest validation loss. T: true class. P: predicted class and probability.

- **Masking test.** The masking test (Supplementary F25) shows that models trained on full images depend on the head. With the background set to zero, every model keeps most of its accuracy, between 93.7% and 96.8% in binary detection and between 94.2% and 99.6% in subtype classification. With the head set to zero, binary detection accuracy falls to between 44.4% and 53.3% for every model except InceptionV3 (65.7%), close to or below the majority-class rate of 62.6%, and subtype classification accuracy falls to between 11.6% and 33.8%, below its majority-class rate of 68.9%.

- **Shortcut test.** The shortcut test shows that this is not the whole story. When each backbone is retrained on background-only images, seven of the nine binary detection models exceed the majority-class rate after Holm correction, reaching 83.0% for Swin-Base, 81.3% for ResNet50 and 77.8% for EfficientNetB0, while MobileNetV3-Large (67.8%) and ViT-B/16 (65.5%) are not significant. In subtype classification all nine are significant, with background-only accuracies between 82.2% and 97.8% (Supplementary T9, Supplementary F26). The background, borders and framing of these public collections therefore carry substantial label information, in the subtype classification pool almost enough to solve the test set without any brain. The models trained on full images do not need the background to perform well, but the data do not prevent them from using it, and the accuracies reported here cannot be taken as proof that pathology has been learned.

4.8. Comparison with Existing Literature

Table 12 places the fold-ensemble results beside published figures for the same or related architectures. The comparison is for orientation only: the published subtype classification studies use the 3,064-image Cheng collection without deduplication, the binary study separates normal from abnormal images of another collection, whereas the

results here are measured after removing near-duplicates and splitting by group, which removes exactly the images most likely to have inflated a score (Kapoor and Narayanan, 2023; Roberts et al., 2021). The subtype classification result rests on 45 test images, and the models here share one training configuration rather than being tuned individually.

Table 12
Present results beside previously published studies. For this work, the dataset size is the deduplicated pool and accuracy is fold-ensemble accuracy on its held-out test set (171 binary detection and 45 subtype classification images).

Study	Model	Dataset size	Task	Accuracy
Talo et al. (2019)	ResNet34	613	Normal vs abnormal	100.00%
Rehman et al. (2020)	VGG16	3,064	Subtype classification	98.69%
Noreen et al. (2020)	DenseNet201	3,064	Subtype classification	99.51%
Noreen et al. (2020)	Inception-v3	3,064	Subtype classification	99.34%
Tummala et al. (2022)	ViT ensemble	3,064	Subtype classification	98.70%
Our work	ResNet50	1,185	Binary detection	98.25%
	EfficientNetB0	1,185	Binary detection	99.42%
	VGG16	4,730	Subtype classification	97.78%
	DenseNet121	4,730	Subtype classification	100.00%
	InceptionV3	4,730	Subtype classification	100.00%
	ViT-B/16	4,730	Subtype classification	100.00%

Three conclusions about architecture can be drawn from these experiments, and one cannot. First, under an equal optimisation budget every backbone family reaches 97% to 99% binary detection accuracy, and the differences between families are not significant on 171 test images, so this benchmark gives no support to the claim that transformers, hybrids or convolutional networks are better suited to the task. Second, GFLOPs differ 72-fold and parameter counts 33-fold for statistically indistinguishable accuracy, which makes cost the deciding factor in practice: EfficientNetB0 matches the leaders with 4.01 million parameters and 0.77 GFLOPs and has the second-highest throughput, which is why it is the architecture deployed in the accompanying application. Third, cross-validation spread is as large as the gaps between models, so single-run leaderboards on collections of this size cannot rank architectures. What cannot be concluded is that any of these models has learned tumour morphology rather than properties of the source collections, since background-only models reach 83% binary detection and 98% subtype classification accuracy.

5. Conclusion

This study benchmarked nine ImageNet-pretrained backbones for binary detection and subtype classification of brain tumours with one pipeline: label-aware perceptual-hash deduplication and group-aware splitting before any training, an equal number of optimiser steps for every model, five-fold cross-validation with fold ensembles, bootstrap confidence intervals and Holm-corrected McNemar tests, calibration analysis, measured efficiency, and attention and shortcut audits for every model. Deduplication removed 63.57% of the binary detection and 18.84% of the subtype classification images, and the leakage audit found no test image near a development image. Decoding, hashing and cropping ran at about 83 to 86 images per second on four CPU cores, and the deduplication itself took a few seconds, so curating pools of this size costs little. Its exhaustive pairwise comparison grows with the square of the pool size, however, and its Hamming threshold of 5, which is equivalent to 4 for these hashes, was carried over rather than validated. On 171 binary detection test images, CoAtNet-0, EfficientNetB0 and ViT-B/16 reached 99.42% accuracy, and no model differed significantly from any other. The subtype classification split was dominated by one group of 1,431 images, so its test set of 45 images cannot rank the models. Temperature scaling fitted on near-separable validation folds made calibration worse for most models. Most importantly, models retrained on background-only images exceeded chance for 7 of 9 binary detection and all 9 subtype classification backbones, with up to 83.04% and 97.78% accuracy, so the public collections used here and in much of the literature carry label information outside the brain, and high accuracy on them does not establish that pathology has been learned.

The study has six principal limitations. The images are public two-dimensional slices of unknown MRI sequence without patient identifiers, so grouping by image similarity is only a proxy for a patient-level split. The binary detection negative class is tumour-negative rather than certified normal. No external validation set was available. Every model

was trained once per fold with one seed and one shared configuration. The deduplication threshold was not validated against pairs confirmed as duplicates, and its effect on the test results was not measured. The processing time of the curation stage was measured on two pools of a few thousand images on one machine, and the hash was not timed on its own. Future work should run the threshold sensitivity analysis of Section 3.2.3, split by patient where identifiers exist, use a smaller grouping tolerance or a different grouping criterion for the subtype classification pool so that its test set represents every class, remove or standardise the image border before training and repeat the shortcut test, fit calibration on larger held-out sets, and validate externally on multi-sequence and three-dimensional institutional data before any clinical claim is made.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All datasets used in this study are publicly available:

- Brain MRI Images for Brain Tumor Detection: https://www.kaggle.com/datasets/navoneel/brain-mri-images-for-brain-tumor-detection
- Br35H: Brain Tumor Detection 2020: https://doi.org/10.21227/tbkk-q937, also at https://www.kaggle.com/datasets/ahmedhamada0/brain-tumor-detection
- Brain Tumor Classification (MRI): https://doi.org/10.34740/KAGGLE/DSV/1183165
- Brain tumor dataset of Cheng: https://doi.org/10.6084/m9.figshare.1512427.v5, in the JPEG conversion used here at https://www.kaggle.com/datasets/moundir8/jun-cheng-brain-tumor-mri-dataset-jpg

The notebooks for both tasks, which perform deduplication, splitting, training, evaluation and figure generation, are available with their outputs (the split of every image, per-run metrics, plots and the trained checkpoints) at a public repository, withheld for anonymous review.

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